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Review

Proteomics in human Parkinson's disease research

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ABSTRACT

During the last decades, considerable advances in the understanding of specific mechanisms underlying neurodegeneration in Parkinson's disease have been achieved, yet neither definite etiology nor unifying sequence of molecular events has been formally established. Current unmet needs in Parkinson's disease research include exploring new hypotheses regarding disease susceptibility, occurrence and progression, identifying reliable diagnostic, prognostic and therapeutic biomarkers, and translating basic research into appropriate disease-modifying strategies. The most popular view proposes that Parkinson's disease results from the complex interplay between genetic and environmental factors and mechanisms believed to be at work include oxidative stress, mitochondrial dysfunction, excitotoxicity, iron deposition and inflammation. More recently, a plethora of data has accumulated pinpointing an abnormal processing of the neuronal protein α -synuclein as a pivotal mechanism leading to aggregation, inclusions formation and degeneration. This protein-oriented scenario logically opens the door to the application of proteomic strategies to this field of research. We here review the current literature on proteomics applied to Parkinson's disease research, with particular emphasis on pathogenesis of sporadic Parkinson's disease in humans. We propose the view that Parkinson's disease may be an acquired or genetically-determined brain proteinopathy involving an abnormal processing of several, rather than individual neuronal proteins, and discuss some pre-analytical and analytical developments in proteomics that may help in verifying this concept.

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Abbreviations: ALS, amyotrophic lateral sclerosis; α -SYN, α -synuclein; AD, Alzheimer's disease; CNS, central nervous system; CSF, cerebrospinal fluid; DA, dopamine; DUB, deubiquinating enzyme; 2-DE, two-dimensional electrophoresis; HE, hematoxylin–eosin; LB, Lewy body; LC, locus ceruleus; LCM, laser capture microdissection; LD, levodopa; LN, Lewy neuritis; LP, lumbar puncture; MPTP, 1-methyl 4-phenyl 1,2,3,6-tetrahydropyridine; NM, neuromelanin; PD, Parkinson's disease; PTMs, post-translational modifications; ROS, reactive oxygen species; SNpc, substantia nigra pars compacta; TMT, tandem mass tags; UB, ubiquitin; UPS, ubiquitin–proteasome system; VTA, ventral tegmental area.

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1. Introduction

Parkinson’s disease (PD) is the most common neurodegenerative movement disorder, affecting 1–2% of the population over the age of 60 years [1] and over 6 million people worldwide. Recent epidemiological projections herald a continuously increasing prevalence along with populations’ aging and it is anticipated that the number of affected individuals may double in the most populous nations within the next two decades [2]. As a result, PD represents a growing socio-economic burden on societies and a major public health challenge for the future, the more so as, despite extensive research, neither the cause nor the mechanisms underlying the condition have been firmly established thus far. Consequences resulting from this uncertainty include poor diagnostic accuracy, absence of valid biomarkers and difficulties to elaborate therapeutic interventions that might significantly influence disease progression. In this review article, we make the assumption that modern proteomic technologies may offer new insights into the pathogenesis of PD and, while broad reviews have already been published in this field [3–5], we will focus on proteomics as a tool to approach pathogenesis of sporadic PD in humans.

2. Neuropathology of Parkinson’s disease: the enigma of progression

Since being formally recognized as a distinct morbid entity in 1817 by James Parkinson in his seminal “Essay on a Shaking Palsy” [6], PD has been essentially regarded as a disorder of motor function, producing a variable, asymmetric and progressive combination of resting tremor, bradykinesia, rigidity

and postural instability. Since the pioneering work of Fritz Heinrich Lewy in 1912 and Constantin Trétiakoff in 1919, the neuropathological lesions at the basis of typical, sporadic PD have been extensively studied and comprehensively defined [7,8]. Classically, there is a progressive loss of neurons in selected pigmented and non-pigmented brainstem and cerebral nuclei, particularly in the substantia nigra pars compacta (SNpc) in the midbrain and the locus ceruleus (LC) in the pons. There is also a mild degree of astrocytic and microglial reaction. Remaining neurons show inclusions, known as pale bodies and Lewy bodies (LBs, Fig. 1a) when located in their cytoplasm, and Lewy neuritis (LNs, Fig. 1d) in their neuronal processes, which have been shown using immunohistochemistry techniques to be notably composed of α -synuclein (α -SYN) (Fig. 1b, f, h) [9]. The temporal and topographical progression of PD pathology in the central (CNS) and, to a lesser extent, the peripheral autonomic nervous system, has been recently refined [10–13]. According to these studies, PD pathology, defined as α -SYN positive aggregates, appears to follow a non-random, caudo-rostral progression within the CNS, starting in the dorsal motor nucleus of the vagus nerve in the brainstem, in the olfactory bulb and anterior olfactory nucleus. Lesions then become more and more pronounced in affected areas while progressing rostrally and involving subsequent structures in a stereotyped and selective manner through what Braak and colleagues have defined as six neuropathological stages of PD, the first two stages corresponding to presymptomatic PD and the next four to symptomatic PD. During this pathological process, selective structures of the brainstem, temporal mesocortex and eventually neocortex are progressively involved. Consequently, involvement of the SNpc, which occurs during PD’s Braak stage 3, is just another step in a much larger, multisystem

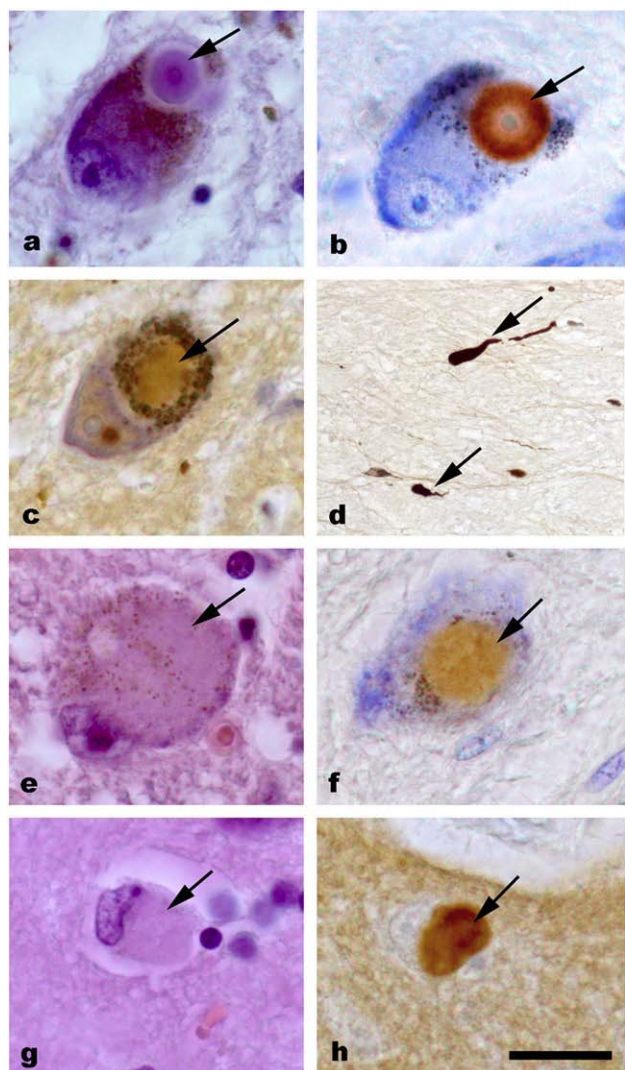


Fig. 1 – Histological lesions of Parkinson's disease: “classical” or “brain stem-type” Lewy bodies in the substantia nigra pars compacta (a–c); Lewy neurites (d); pale bodies (e and f) and cortical Lewy bodies in the entorhinal cortex (g and h). (a, e and g: HE staining; b, d, f and h: immunohistochemistry with antibody anti- α -synuclein; c: immunohistochemistry with antibody anti-ubiquitin. Scale bar: a–c and e–h: 20 μ m, d: 60 μ m).

pathology, which involves not only motor but also non-motor, autonomic, limbic and cognitive systems. It is important to appreciate that, in this scenario, neurodegeneration is initiated a considerable time before symptoms become detectable — probably 5 to 10 years — and the disease clinically starts when about 50 to 70% of nigral neurons have already been lost [14]. Thus, an early clinical diagnosis of PD is necessarily made at an advanced pathological stage.

However, the concept proposed above has been recently challenged mostly because of the lack of correlation between the Braak staging system and the clinical severity of PD, casting doubt about its predictive validity [15,16]. As an alternative, it has been hypothesized that the regional pattern of α -SYN deposition may reflect the relative propensity of

specific brain regions to exhibit Lewy pathology, irrespective of the timing of events. According to this view, there is no caudo-rostral progression and the common involvement of the SNpc, the LC and the dorsal motor nucleus of the vagus nerve, either alone or in various combinations, reflects the individual likelihood and vulnerability of these structures to develop Lewy deposits.

From a research point of view, whatever the mode of progression, these neuropathological considerations imply that information extracted from human samples, taken at a certain time point during PD progression, will, somewhat artifactually, offer an instantaneous view of a constantly evolving pathology and may not necessarily reflect the complex dynamics of the whole degenerative process.

3. Selective neurodegeneration

Whereas the etiology of sporadic PD remains unknown, many pathogenic mechanisms have been shown to be likely at work in the SNpc of PD patients, including oxidative stress and production of reactive oxygen species (ROS) in excess, mitochondrial dysfunction and energy production imbalance, excitotoxicity mediated by increased glutamatergic stimulation, excessive deposition of reactive iron and abnormal iron-neuromelanin (NM) interaction, inflammation and immune-mediated reactions, defects in neurotrophic factors, and apoptosis, among others [17]. More recently, an abnormal cellular protein processing and dysfunction of the ubiquitin-proteasome system (UPS) have been suggested to be central to LB formation and neurodegeneration. It is likely that more than one of these mechanisms is involved, yet the exact combination and succession of which remain unclear.

The nigrostriatal pathway is formed by DA-containing neurons projecting to the striatum (i.e., putamen and caudate nucleus) whose cell bodies are confined in a densely populated SN subdivision individualized as the pars compacta. The SNpc acts as a modulator of motor function via a dual effect upon the basal ganglia loops, through dopamine receptor D1-mediated activation along the direct pathway and D2-mediated inactivation of the indirect pathway, to promote voluntary and inhibit involuntary movements. In PD, the extensive neuronal degeneration of this dopaminergic pathway alters the balance between both loops resulting in the typical features of PD, in particular bradykinesia. While PD neurodegeneration is not restricted to dopaminergic systems, the reasons why a selected population of DA neurons is primarily targeted while others remain marginally or not at all affected, is an important issue to be addressed. Indeed, neurodegeneration of the DA neurons does not seem to occur randomly or diffusely within the SNpc or the other DA systems. In fact, nigral lesions are regionally highly selective with the ventrolateral and caudal tiers being primarily and more severely affected, whereas the dorsomedial tier is more affected during normal aging [18]. Although age constitutes the most important risk factor for the development of PD, the processes inducing age-related dopaminergic neuronal loss seem to be anatomically distinct from those underlying PD. This heterogeneity and selective vulnerability of DA — and also non-DA — neurons has been hypothesized to result from morphological specificities. Indeed, a relatively low number of

DA neurons (about 450,000 in adult human SNpc) provide massive striatal innervations through disproportionately long, thin and branched axonal projections, compared to a relatively small cell body, which are poorly or not myelinated. This type of neuron may need higher energy production to achieve basic functions, particularly axonal transmission, and may be more vulnerable to any form of insult [19]. Compared to most brain structures, DA neurons are more exposed than others to oxidative stress. Their elevated rate of oxygen and calcium metabolism, low levels of antioxidants and high iron content are all factors inducing higher rate of damaging ROS. Thus, their specific physiology may also hold the key of their vulnerability. For example, DA neurons maintain an autonomous activity relying on L-type calcium channels, whereas other neurons use sodium channels [20]. This reliance implies an increased and sustained energy consumption to sequester and pump calcium out of the cell. Ultimately, this imposes an increased metabolic stress on mitochondria, accelerating cellular ageing and death [20]. In addition, DA itself may predispose, induce or facilitate degeneration. Its chemical instability and auto-oxidation capability, occurring both spontaneously and enzymatically, may potentially lead to ROS production. Two populations of DA neurons can be observed in the mesencephalon, those containing NM, a major DA oxidation by-product [21], and those without. A correlation between the vulnerability of nigral neurons and NM pigmentation has been established. Indeed, NM-containing neurons which comprise not only DA neurons of the SNpc but also noradrenergic neurons of the LC seem to be more prone to degenerate in PD [22]. Thus, not only DA but also NM itself could be cytotoxic and contribute to oxidative stress through its interaction with iron and other reactive metabolites [22].

Among DA neurons, another pattern of regional vulnerability has been observed, with some populations showing less or no loss at all [23–25]. For example, adjacent midbrain DA populations within the ventral tegmental area (VTA) and their projections to the ventral striatum exhibit a relative resistance to PD-related degeneration. In contrast to SNpc, VTA neurons contain little NM, indicating less DA-related oxidative stress during neuron life span [26]. Furthermore, it is also possible that neuron surrounding influences their fate critically. Indeed, susceptible neurons tend to lie in calbindin-poor regions of the SNpc [23], known as nigrosomes, whereas neurons in calbindin-enriched regions, known as the nigral matrix and containing neuron terminals issued from the striatum, are spared. The significance of this topographical distinction is unknown. Finally, and at variance with the observations supporting the neuron body as the main target of degeneration in PD, it has also been proposed that the loss of nigral neurons may result from a “dying back” process whereby striatal nerve terminals are the primary target of the degenerative process [27]. Indeed, the degree of terminal loss in the striatum appears to be disproportionately more pronounced compared to the severity of SNpc neuron loss [28,29]. Whereas this scenario may explain the topology of neuronal loss in the SNpc which is somatotopically connected to the putamen, it seems at odds with the notion of structural compensatory mechanisms that are believed to occur in the striatum to attenuate the effects of reduced DA transmission, one of which relying on synaptogenesis, i.e. sprouting of new terminals by surviving neurons [30].

Altogether, neurodegeneration of PD does not occur at random and appears to be topographically patterned not only throughout the CNS but also within affected structures. Understanding the basis of this selectivity, which is likely to somehow reflect pathogenesis [23], may be of crucial importance.

4. Lewy body

Lewy inclusions, the pathological hallmark of PD, are found in the cytoplasm of surviving neurons of the SNpc but also in non-dopaminergic neurons such as the LC or the cortex. They exhibit a variable morphology, developing as isolate or multiple, spherical or elongated, 8 to 30 μm in diameter, LBs in cell bodies, or spindle-like LNs in neuronal processes. Composition of LBs, molecular mechanisms of aggregation, biological role and implication in the pathogenic processes underlying PD are still unclear, in spite of the identification of α -SYN as an important component [9]. Importantly, α -SYN inclusions are not specific for PD and occur also in other neurological conditions, tentatively classified as “alpha-synucleinopathies”, which include besides PD, dementia with LBs, the LB variant of Alzheimer’s disease (AD), multiple system atrophy, and neurodegeneration with brain iron accumulation type I, among others.

4.1. Structure and composition

In PD, two variants of intracytoplasmic LBs can be distinguished: the classical brainstem type found within the perikaryon of DA nigral neurons and a less common type found in cortical neurons. At light microscopy using hematoxylin/eosin (HE) staining, classical LBs typically exhibit a dense core surrounded by a clearer halo (Fig. 1a). Cortical LBs are less eosinophilic and present with a more homogeneous aspect lacking the distinctive core and halo (Fig. 1g). The greater sensitivity of immunostaining analysis has allowed an easier identification of LBs [31] and demonstrated that α -SYN labels specifically the halo whereas ubiquitin (UB) staining yields a more diffuse labelling (Fig. 1b, c). Other structures weakly stained with HE but clearly positive for α -SYN and UB are also detected, including pale bodies (Fig. 1e, f). Some data suggest that pale bodies correspond to LBs at an early stage of development [32,33]. Ultrastructurally, classical LBs seem to be composed of fibrils radiating out from a central core of dense granular material whereas the cortical type is composed almost exclusively of circular or oval material [34]. Our knowledge regarding the biochemical composition of LBs has evolved with more sophisticated immunohistochemical techniques, showing that LBs were notably positive for many compounds besides UB and α -SYN. In fact, the use of various non-proteomic techniques has led to a list of nearly 80 proteins identified thus far within LBs (Table 1).

4.2. Alpha-synuclein

α -SYN belongs to the synuclein family of proteins composed of α , β and γ -synuclein, which are extensively expressed in the brain. Although the exact physiological functions of α -SYN remain uncertain, it may be involved in neurotransmission by regulating synaptic vesicles size, recycling and plasticity [35]. α -SYN is a

Table 1 – List of proteins identified in Lewy bodies using non-proteomic, essentially immunohistochemistry staining techniques.

Advanced glycation endproducts	Complement proteins (C3d, C4d, C7, C9)	Immunoglobulins (IgGs)	P35 ^{nck5a}	Synaptic vesicle-specific protein
α -crystallin	Chondroitin sulfate proteoglycans	Phosphorylated I κ B α	P38	Synaptophysine
α-synuclein	Chromogranin A	Heme oxygenase-1	p62/sequestosome 1	Synaptotagmine
α 2-macroglobulin	Cullin-1	Histone deacetylase 4	Parkin	Tau
Agrin	Cyclin B	Leucine-rich repeat kinase 2 (LRRK-2)	Phospholipase C- δ	Tissue transglutaminase
Amyloid precursor protein (APP)	Cyclin-dependent kinase 5	Lipids	PTEN-induced kinase 1 (PINK1)	Torsin A
Apolipoprotein J/clusterin	Cytochrome c	Microtubule-associated proteins (MAP-1, 1b, 2, 5)	Prolyl-isomerase Pin 1	Tropomyosine
26S proteasome ATPase	DJ-1	Multicatalytic proteinase	Retinoblastoma protein	Tubulin
Basic fibroblast growth factor	Dorfin	mxA protein	ROC1	Tubulin polymerization promoting protein/p25
β -TrCP	Extracellular signal-regulated kinases	NEDD8	sept4/H5	Tyrosin hydroxylase
C terminus of Hsp70-interacting protein	14-3-3 protein	Neurofilaments	SIAH-1	Ubiquitin
Calcium/calmodulin-dependent protein kinase II	G-protein-coupled receptor kinase 5	NF κ B, I κ B α	Sphingomyelin	Ubiquitin activating enzyme (E1)
Calbindin D _{28K}	Gelsoline-related amyloid protein Finnish type	NUB1	Cu/Zn superoxide dismutase	Ubiquitin conjugating enzyme UbcH7 (E2)
Centrosome/aggresome-related proteins	Glyceraldehyde-3-phosphate dehydrogenase	omi/HtrA2	Mn superoxide dismutase	Ubiquitin C-terminal hydrolase (UCH-L1)
CHIP	Heat shock proteins (HSP 27, 40, 60, 70, 90, 110)	pael-R	Synphilin-1	Vesicular monoamine transporter 2

According to Shults CK, 2006, and Wakabayashi K et al., 2007.

Genetic defects in proteins highlighted in bold have been reported to cause or increase the susceptibility for Parkinson's disease.

highly conserved 140-aminoacid presynaptic protein of ~19 kDa originally identified in humans as the non-amyloid beta component (NAC) of AD amyloid plaques [36]. Its structure can be divided into three regions. First, a 60-aminoacid N-terminal region containing 11 imperfect repeats with highly conserved KTEGV motifs, is predicted to form amphipathic alpha-helix typical of apolipoprotein-like-class-A2 which mediates binding to phospholipid vesicles [37]. Second, a central region containing the NAC sequence from residue 61 to 95 which appears to be essential for aggregation [36]. Third, a C-terminal portion, less conserved, containing many acidic residues which confer to the molecule a low pI. Natively, α -SYN presents in a soluble, unfolded state in the cytoplasm but can be found in more ordered states such as α -helical conformation in the presence of lipid membranes or as β -sheets in fibrils composing LBs [38,39].

Involvement of α -SYN in PD pathogenesis derives from the identification of missense mutations (A53T and A30P) in the SNCA gene in early-onset familial PD kindreds with autosomal dominant inheritance [40,41]. Interestingly, predictions suggest that these N-terminally located mutations could disrupt the α -helix and extend the β -sheet structure, which is thought to be involved in the self-aggregation of proteins and the formation of amyloid-like structures. Perhaps more importantly, duplications and triplications of the α -SYN gene SNCA were also reported in autosomal dominant forms of PD, resulting in increased levels of wild-type α -SYN in the absence of other known mutations [42,43]. These data suggest that abnormal intracellular accumulation of wild-type α -SYN is sufficient to induce neurodegeneration, a

mechanism that might be common to sporadic PD as well, in view of the recent finding that some polymorphisms of the promoter region of the SNCA gene may increase the risk of PD in the general population [44–46]. In addition, many factors that can promote α -SYN aggregation have been identified. For example, α -SYN was shown to carry post-translational modifications (PTMs) such as phosphorylation (at Ser129) and ubiquitination (at Lys12, Lys21 and Lys23) that could contribute to its toxicity [47]. Deleterious interactions between α -SYN and environmental agents or other proteins could also dramatically accelerate the rate of α -SYN fibrillogenesis in vitro and impairing the process of degradation of the abnormal protein and/or increase oxidative stress [48,49].

5. Potential mechanisms underlying degeneration

5.1. Oxidative stress and mitochondrial dysfunction

Post-mortem studies have consistently implicated oxidative damage to lipids, proteins, and DNA in the pathogenesis of PD, as observed in the SNpc of sporadic PD brains [50]. The source of this increased oxidative stress is uncertain but may include mitochondrial dysfunction, increased dopamine metabolism, reactive iron deposition and impaired antioxidant pathways. In fact, much evidence suggests a major role for defects in complex I of the respiratory chain leading to decreased ATP production as well as damage caused by excess production of

ROS. Consistent with this hypothesis, altered complex I activity was observed in the SNpc of sporadic PD patients [51]. In addition, several observational and epidemiological studies suggest that pesticides and other environmental toxins that inhibit complex I are involved in the pathogenesis of sporadic PD [52]. For example, 1-methyl 4-phenyl 1,2,3,6-tetrahydropyridine (MPTP) was initially identified as an accidental by-product during the illicit manufacture of a synthetic opiate in Northern California in the early 80s, and drug users who injected MPTP developed an acute syndrome closely resembling PD [53]. The selectivity of MPTP for dopaminergic neurons relates to its conversion in astrocytes by monoamine oxidase B to the active metabolite, 1-methyl-4-phenyl pyridinium (MPP⁺), which is taken up in DA neurons through the dopamine transporter, where it inhibits complex I and ultimately leads to cell death [27]. Similarly, the widely used herbicide paraquat, the insecticide rotenone and the fungicide maneb (manganese ethylenepisthiocarbamate) all inhibit complex I and induce a selective loss of nigrostriatal neurons in rodents [54]. In rotenone-treated rats, LB-like intraneuronal filamentous inclusions containing α -SYN and UB have been observed [55]. In addition, a connection between α -SYN and complex I is supported by the observation that α -SYN knockout mice are resistant to MPTP, whereas α -SYN transgenic mice show enhanced toxicity [56], suggesting that α -SYN is required for mediating the downstream effects of complex I inhibition. These findings are consistent with alterations in complex I being central to the pathogenesis of sporadic PD and support environmental factors as main contributors to PD pathogenesis. However, the role of mitochondrial defects and oxidative stress has also been challenged by several authors arguing that these alterations may merely reflect the unspecific and additional effects of aging [57] or the consequences, rather than the cause, of neuronal dysfunction.

5.2. Protein aggregation and the ubiquitin–proteasome system

Aggregation of soluble proteins into insoluble complexes and inclusions is now considered at the core of many neurodegenerative conditions, including PD. One pivotal mechanism for proteins to aggregate relies on the exposure of hydrophobic segments that are prone to form inappropriate intermolecular binding, when proteins are unfolded, partially folded or misfolded. It is noteworthy, however, that protein aggregation does not seem to occur at random but is a highly specific phenomenon occurring between similar hydrophobic proteins or distinct proteins sharing a common aggregation-promoting motif [58]. During this process, many native proteins not primarily involved in the aggregation formation, may be recruited and trapped into growing inclusions, as “innocent by-standers”.

Under physiological conditions, neurons handle abnormal or altered (genetically modified, denatured, misfolded, etc) cellular proteins by distinct mechanisms, notably molecular chaperone-mediated refolding, degradation by the ubiquitin–proteasome system (UPS) and peptidases, and, when the production of abnormal polypeptides exceeds the capacity of the UPS, sequestration into microtubule-dependent aggre-

somes. Under non-physiological conditions, such as molecular crowding, oxidative stress, PTMs or modifications of the protein primary structure, soluble monomers and oligomers of native or non-native polypeptides may self-assemble, polymerize and coalesce into giant intracellular inclusions, such as LBs in PD [59]. In fact, it has been suggested, based on many similarities between both structures, that aggresomes and LBs may be created following the same cytoprotective process of protein sequestration. Whereas aggresomes may be cleared by enhanced proteasome activity and autophagy, LBs may not, for reasons that are yet unclear [60]. Eventually, LBs and LNs may impair critical intracellular processes, notably axonal transport, synaptic plasticity and neurotransmission, and eventually promote degeneration.

Because the UPS is a major pathway mediating the degradation of abnormal cellular proteins [61], it has been proposed that its dysfunction may be a key factor in the pathogenesis of PD and LB formation (reviewed in [62]) either primarily or secondary to an impairment of the capacity of the UPS to handle an overwhelming quantity of altered proteins. In particular, failure of the 26S proteasome may be critical [63]. For example, UB has been observed to concentrate in LBs where it colocalizes with chaperones and UPS components that normally take part in the protein quality control machinery and clearance of abnormal proteins in the cell. To further support this view, it has been recently shown that systemic administration of the proteasome inhibitors epoxomicin or PSI in rats induces a gradually progressive, levodopa (LD)-responsive, form of parkinsonism similar to PD in humans [64]. The central role of the UPS in PD pathogenesis is, however, still a matter of disagreement [65].

5.3. Many unsolved issues

The above review of the potential mechanisms underlying the pathogenesis of PD shows that important issues remain unsolved. First, since 1997, a plethora of genetic defects has been reported in familial forms of PD, with an autosomal dominant or recessive mode of inheritance. To date, nearly twenty loci have been implicated, from which ten different causal as well as susceptibility gene mutations have been formally identified (Table 2). In general, with the notable exception of PARK9, PARK14 and PARK15, the phenotype of genetically-determined PD is quite similar to sporadic PD, but age at onset is usually younger. The involvement of some of these genes tends to support the mechanisms of degeneration described above. For example, PARK2 is characterized by genetic defects in the protein parkin which acts mainly as an E3 ubiquitin ligase essential for the UPS. Similarly, PARK6 results from mutated forms of PTEN-induced kinase 1, a mitochondrial protein that seems to have a protective role against proteasome inhibitors. On the other hand, pathogenic mechanisms associated with PARK8 and more recent gene abnormalities fit less easily into current theories. Although information obtained through genetic approaches has proved immensely useful to elaborate new hypotheses regarding the pathogenesis of sporadic PD, the diversity of genes uncovered so far makes it difficult to establish a unifying hypothesis that takes all or most known gene products into account. Second, it is still not clear whether LBs and LNs are the toxic entities promoting

Table 2 – PARK-assigned causal and susceptibility genetic defects leading or contributing to Parkinson's disease.

Locus ^a	Chromosome location ^a	Gene symbol ^a	Mutation	Mode of inheritance	Gene product	Protein function	Age of onset	Features	Lewy bodies
PARK1	4q21.3-q22	SNCA	A53T, A30P, E46K	AD (sometimes in sporadic cases)	α -synuclein	Synaptic protein of unknown function	45–60 years	Early-onset PD with early dementia	Yes
PARK2	6q25.2-q27	PARK2	Point mutations, deletions, multiplications	AR (rarely AD)	Parkin	E3 ubiquitin ligase	<20 years	Early-onset PD with slow progression	A few
PARK3	2p13	PARK3	?	AD	?	?	60 years	Late-onset typical PD	Yes
PARK4	4q21.3-q22	SNCA	Duplication, triplication	AD	α -synuclein	Synaptic protein of unknown function	45–60 years	Early-onset PD with early dementia	Yes
PARK5	4p13	UCHL1	Mutation	AD	Ubiquitin carboxylterminal hydrolase L1	Deubiquitylating enzyme	45–60 years	Early-onset typical PD	?
PARK6	1p36.12	PINK1	Missense mutations, truncations	AR (rarely AD)	PTEN-induced kinase 1	Mitochondrial protein of unknown function	20–45 years	Early-onset PD with slow progression	Yes
PARK7	1p36	DJ-1	Deletions, truncation, missense mutation	AR	DJ-1 protein	Astrocytic and neuronal protein of unknown function	20–45 years	Early-onset PD with slow progression	?
PARK8	12q12	LRKK2	G2019S, I2020T, missense mutations	AD (sometimes in sporadic cases)	Leucine-rich repeat serine/threonine protein kinase 2	Protein kinase	>50 years	Late-onset typical PD	Usually yes
PARK9	1p36	ATP13A2	Missense mutations	AR	ATPase type 13A2	Lysosomal type 5 Ptype ATPase	<20 years	Atypical (Kufor-Rakeb disease)	?
PARK10	1p32	PARK10	?	Susceptibility	?	?	60–70 years	Typical PD	?
PARK11	2q37.1	GIGYF2	Missense mutations	AD	GRB10-interacting GYF protein 2	Brain protein of unknown function	60–70 years	Typical PD	?
PARK12	Xq21-q25	PARK12	?	Susceptibility	?	?	60–70 years	Typical	PD
PARK13	2p13.1	HTRA2	Missense mutations, polymorphism	Susceptibility	Omi/HtrA serine peptidase 2	Mitochondrial serine protease	60–70 years	Typical PD	?
PARK14	22q13.1	PLA2G6	Missense mutations	AR	Phospholipase A2, group VI	Fatty acid release from phospholipids	10–25 years	Atypical (dystoniaparkinsonism)	?
PARK15	22q11.2-qter	FBX07	Missense, truncating, compound mutations	AR	F-box protein 7	Molecular scaffold in the formation of protein complexes	10–20 years	Atypical (pallidopyramidal disease)	?

^a According to www.genenames.org (March 2009).

degeneration by impairing the normal functioning of neurons and axons, or whether it represents a protective mechanism of defence by which neurons sequester altered proteins to limit their toxic effects upon neuronal homeostasis [66]. In the latter scenario, degeneration occurs late in the process, when the capacity of LBs to circumvent toxic compounds is overwhelmed. A major argument in favor of a protecting role of LBs in PD comes from the observation that PARK2- or PARK8-affected individuals do not consistently show LBs, suggesting that LBs are not necessary for nigral degeneration to occur. The neurotoxicity of elevated levels of various α -SYN intermediates in the aggregation process (monomers, oligomers, protofibrils and fibrils) has also been extensively studied (recently reviewed in [67–69]) and each has been suspected to be the triggering agent of degeneration in PD, rather than mature LBs. This issue is crucial with respect to developing neuroprotective agents, because interventions aiming at interfering with protein aggregation may turn out to be deleterious by inhibiting the formation of potentially protective LBs or by favoring the accumulation of toxic intermediates. Third, the temporal and mechanistical relationships between oxidative stress, mitochondrial dysfunction and protein aggregation are still unclear and, perhaps more importantly, the promoters, i.e. the causal factors, of these abnormal biological processes are completely unknown [70,71]. Fourth, the selectivity of the structures involved in PD, as described above, can hardly be explained by the currently known pathogenic factors likely to operate in the PD brains, and the morphological hypotheses developed by Braak and colleagues are difficult to incorporate into the suspected scenarios associating oxidative stress, mitochondrial dysfunction and proteasome dysfunction.

In order to examine these unanswered, yet fundamental questions, and because PD is now viewed as a disorder of brain protein handling — or a “protein deposition/conformational disease” — we make the assumption that proteomics-based studies of relevant brain samples might be one of the most appropriate and unbiased strategies to approach PD pathogenesis.

6. Proteomics applied to Parkinson's disease research

During the past few years, a number of proteomic studies have been performed in the field of PD research both in animal models of and in humans with PD.

6.1. Proteomics in animal models of PD

Numerous animal models of PD have been designed using species as diversified as the worm *Caenorhabditis elegans*, the fly *Drosophila melanogaster*, various strains of mice and rats, and non-human primates [72]. Neurodegeneration has been obtained by means of various protocols of exposure to the neurotoxins 6-hydroxydopamine, MPTP, paraquat, rotenone or to proinflammatory molecules such as lipopolysaccharides. More recently, genetically-modified animals have been engineered according to the identified gene defects found in humans. While these animal models have proved very useful to increase our understanding of the pathophysiology of the basal ganglia, to dissect some mechanisms engaged in the

degeneration of DA neurons and to establish the basis for future neuroprotective therapies, they also demonstrated a number of limitations. At present, none of them consistently recapitulate all clinical and neuropathological features of human sporadic PD. Moreover, it has been particularly difficult to model the progressive time course of PD, to replicate the pattern of neurodegeneration involving selected nigral and extra-nigral brain structures over time and to produce typical LBs [73]. While almost each group working on PD basic research has been able to produce its own animal model of PD, the gold standard in this field is still awaited. Proteomics technologies have been applied to some of these animal models [74–87], and these aspects — which extend beyond the scope of the present article devoted to human sporadic PD — have been extensively reviewed in [88] to which the reader is referred for more detailed information.

6.2. Proteomics in human PD

In contrast to animal models which can be quite easily made available as needed, the type, number, quantity and quality of samples that can be obtained from humans are limited. However, samples from neurologically-affected patients are of enormous interest because they may show abnormalities directly linked to the condition under study. In PD, as in other neurological diseases, most studies have relied on samples of cerebrospinal fluid (CSF) obtained from living patients and of brain tissue obtained at autopsy, the value of each being dependent upon the proposed aims of the study. Both types of sample have advantages and limitations that require to be recognized when interpreting data.

6.2.1. CSF protein profiling

CSF may represent an ideal source of diagnostic, prognostic of therapeutic biomarkers, because of its proximity to brain structures undergoing degeneration which may release pathogenically relevant molecules into the extracellular space. In addition, CSF is available over patient's lifetime which makes possible the study of protein profile changes during the entire disease course. Indeed, following the introduction and routine use of atraumatic needles in the clinic, lumbar puncture (LP) is now considered a minimally invasive procedure that can be performed at any time during PD progression and repeated in the same individual. The quantity of CSF that can be obtained from a single LP is between 5 and 20 ml, which corresponds roughly to 2 to 8 mg of proteins, a figure that allows most proteomic technologies to be performed in multiplicates.

Proteomic approaches of CSF have already proved successful for other neurodegenerative conditions such as AD [89] and Creutzfeldt-Jakob disease [90]. Extensive CSF 2-DE maps have been published [91,92] as well as using other separation methods [93,94] and, thus far, over 2500 proteins have been identified in human CSF [95]. Advantages and limitations regarding the use of CSF in proteomics have been discussed elsewhere, in particular with respect to sample preparation which requires specific handling [96,97]. One of the major issues relies on the high dynamic range of protein concentrations contained in CSF, a small number of which overwhelming a multitude of low abundance, yet of key value, proteins. This problem may be solved by pre-analytical fractionation or immunoaffinity techniques allowing the few more abundant proteins to be removed.

In PD, because degeneration affects only a limited subset of brain structures, it is anticipated that potential biomarkers will be found as low or very low abundance proteins, as demonstrated, for example, by the CSF determination of the monomeric form of α -SYN which has proved difficult because of its minute concentration in this biological fluid. Recently, a group firmly established the measurement of α -SYN in plasma and CSF [98]. On average, a decreased concentration of α -SYN was observed in the CSF of PD patients compared with age-matched controls and this decrease correlated with disease severity, although values from both groups were largely overlapping [99]. Moreover, CSF levels of DJ-1 were found to be significantly increased in sporadic PD and well correlated with disease progression [100]. DJ-1 was also found into the serum of both healthy controls and PD patients, but no significant level differences were observed between the two groups [101]. The distinct behavior of DJ-1 levels between CSF and serum could be explained either by a diluting effect of the protein in serum or by a potential oxidation of the protein that may have prevented antibodies directed against native DJ-1 to recognize the post-translationally modified protein. The latter hypothesis is supported by the fact that abnormal oxidation of DJ-1 has indeed been observed in PD patients [102].

Historically, in 1984, Harrington and Merrill studied CSF samples from 20 PD patients by the then available 2-DE technology and were able to detect in 75% of their cases, but in none of 91 healthy controls, the appearance of a single abnormal spot (129 in their nomenclature) present within the Ig light chains train of spots [103]. Quantitative changes of 20 additional spots were also noted, notably apolipoprotein A1 and orosomucoid. Unfortunately, these findings were never confirmed by this group nor replicated by others ever since. More recently, a more advanced study based on a multiplex quantitative platform using iTRAQ was able to detect various changes in the composition of CSF from patients with AD, PD, dementia with LBs and controls [104]. Over 1500 proteins were identified and 72 were reported to be uniquely associated with PD. Ceruloplasmin and apolipoprotein H were confirmed by Western blot analysis and appeared to be good candidate markers for PD. For instance, the deposition of ceruloplasmin, a major iron-carrying protein, had already been found to correlate with disease severity [105] and to be decreased in the blood of PD patients [106]. However, a recent study challenged the significance of this finding, as considerable overlap in individual values between PD patients and controls was observed [107]. In another study, the sera of 422 patients with PD, amyotrophic lateral sclerosis (ALS), related neurological disorders and unaffected individuals were investigated using quantitative 2-DE [108]. In this unbiased approach, differences in the expression of 7 out of 9 complement-related proteins were found. The PD and ALS populations showed a marked increase in components of complement C3 (C3c and C3dg) and complement factor H. Moreover, factor B was more elevated in PD, but not in ALS, with respect to controls. These results showing differences in the pathways of complement, including C3 phosphorylation differences, between ALS and PD provide independent evidence for the role of systemic, immune-mediated inflammatory processes possibly involved in these conditions.

Currently, no single biomarker has been validated for PD and it is possible that a combination of several of them may prove superior. For example, a large scale study using Luminex-based multianalyte profile reported the performance of a panel of eight CSF proteins (tau, amyloid β_{42} , β_2 -microglobulin, vitamin D binding protein, apolipoprotein A-II and apolipoprotein E, brain derived neurotrophic factor, and interleukin-8) to distinguish PD from AD patients and controls. Authors were able to identify effectively PD patients with a 95% sensitivity and 95% specificity [109] and these encouraging results will require confirmation by independent groups.

6.2.2. Brain tissue protein profiling

At variance with the use of samples obtained from living patients, in which only a tentative—probable at the very best—diagnosis of PD can be made, brain samples have the considerable advantage of being assessed by neuropathological examination, providing a definite diagnosis of PD. Regions of interest can be precisely dissected, for example the SNpc, LC or cerebral cortex, thus increasing the study specificity. The large amount of tissue available may allow individual cells, i.e. DA neurons to be separated, or subcellular fractions, i.e. LBs, to be purified. Finally, and more importantly, direct analyses of the very structures that are primarily affected in PD may provide a comprehensive and simultaneous view of the degenerative mechanisms that are likely to be at work. The major theoretical limitation of this approach involves the potential contamination of analyses by post-mortem alterations [110]. In fact, this issue has been recently examined by independent groups using human [111] and mouse [110, 112, 113] brain tissue samples exposed to increasingly long post-mortem interval, up to 72 h. Somewhat surprisingly, these studies consistently showed that, even for prolonged post-mortem delay at room temperature, brain tissue does not undergo massive degradation and only a minority (6.5% of spots after 48 h at room temperature in [111]) of proteins show significant modifications, including dihydropyrimidinase related protein-2, GFAP and, importantly for PD research, synuclein proteins. Most changes occur after 48 h. The problem of artifactual post-mortem changes may thus not be as significant as previously feared and can be circumvented by limiting the post-mortem delay to less than 24 h and by including in the study a control group of neuropathologically intact brain samples with matched post-mortem delay, whenever possible. Lists of proteins vulnerable to post-mortem delay are available in [110] and [111]. Additional issues include the contribution of unspecific age-related changes to the pathology, the potential impact of the many and long-used pharmacological treatments upon protein expression and a selection bias owing to the fact that, with few exceptions, only patients with advanced PD come to autopsy. This last point is important because it may obscure data interpretation especially when assessing changes of protein expression for the purpose of establishing early biomarkers of the condition. Ideally, PD patients with large range of disease durations should be studied.

Fasano et al. were the first to profile the human PD SNpc using 2-DE. This group was able to identify a total of 44 proteins from which nine were found differentially expressed in PD compared to age-matched controls [114]. Of note, there was an increase of some ROS-scavenging and mitochondrial proteins (peroxiredoxin II, mitochondrial complex III or ATP synthase D)

in PD specimens, supporting the role of oxidative stress in PD pathogenesis. Using a similar approach, the SNpc study by Werner et al. led to the detection of 221 spots found differentially expressed between PD and control groups, contrasting with 321 spots showing a strictly conserved pattern of expression. Out of a total of 37 identified proteins, they found 16 differentially regulated, including elements of iron metabolism (ferritin H), glutathione-related redox metabolism (glutathione S-transferase M3, P1, O1) or novel redox proteins such as SH3BGR1 [115]. Importantly, expression of DJ-1 and UCH-L1 was not different in the two groups and twelve of the 16 differentially regulated proteins displayed only modest changes, as the level ratio of the PD over control group was below 1.5 fold. Comparing the data from these two similar studies, 8 common proteins were found yet none of the differentially expressed ones were concordant. More recently, an extensive investigation of the human SNpc using a shotgun proteomic approach combining ICAT or iTRAQ and MS or MS/MS identified a total of 1263 non-redundant proteins, but the complete dataset of protein identifications was not provided [116].

As LB pathology eventually extends from the brainstem to the cortex, efforts were also made to characterize the human frontal cortex in PD [117]. Using LCMS/MS-based technology and iTRAQ labelling, a study reported the characterization of human cortical proteins affected during PD progression [118]. About 800 proteins were identified, and approximately 200 were found to be differentially expressed between PD patients at various stages and controls, including the 70-kDa mitochondrial heat shock protein mortalin which was found to decrease with disease progression. Recently, it has been demonstrated that CRP40, a splice variant of mortalin, exhibits a catecholamine binding function that may confer neuroprotective properties to this chaperone-like protein [119].

6.2.3. Cellular and subcellular proteomes

To allow a deeper probing of the neural proteome, various strategies have to be elaborated to reduce the complexity of the sample prior to analysis and to select only a discrete, but highly relevant subset of the entire proteome [120]. When using PD brain samples, this concept translates into assessing affected tissue and cells with an increasing level of specificity, ranging from the entire SNpc, or other LBs-enriched regions, to DA neurons and eventually to LBs and other relevant subcellular or even molecular fractions.

It is important to appreciate that the SNpc is composed of different cell types among which only 5 to 10% are DA neurons. Thus, a 5-fold change in protein expression in DA neurons will be diluted to 0.5-fold or less and hardly detectable when examining the whole SNpc. Recent improvements in sampling techniques such as the development of laser capture microdissection (LCM) have allowed the separation of cell populations of enhanced purity from heterogeneous tissue sections via direct visualization of the cells of interest [121]. With this technique, a finely-focused laser is used to cut around or capture cells of interest out of fixed tissue slices (reviewed in [122]). Although LCM was commonly applied in genomic molecular profiling, the approach remains limited in proteomics. The main drawback, precluding a more widespread proteomic application, lies in the lack of protein amplification

systems, such as PCR for nucleic acid, as well as the time consuming and cumbersome nature of cell collection. Indeed, the large number of cells to be extracted from a tissue section necessary for 2-DE analyses [123–125] or a shotgun proteomics analysis followed by MS [126,127] may vary from 10,000 to 100,000 cells. These figures may decrease in the near future with the emergence of more sensitive MS techniques and automated methods to collect cells. Other purification methods have been tested for this purpose but it is uncertain whether these techniques are suitable for proteomics [128–130]. Leverenz et al. examined the content of LBs dissected by LCM from neurons of the frontal cortex from patients with dementia with LBs [131]. The analysis of approximately 2500 cortical LBs led to the identification of 296 proteins, of which 17 had been previously associated with brainstem or cortical LBs, including α -SYN. The presence of one protein, the heat shock cognate 71 was validated by immunohistochemistry not only in the brainstem but also in cortical LBs. More recently, another group, using LCMS/MS, studied cortical LBs from patients with the Lewy variant of AD that were enriched by sucrose gradient centrifugation [132]. These researchers were able to identify about 550 proteins from which 40 were found to co-purify with LBs including some well known LB-associated proteins such as α -SYN, UB, UCH-L1 and proteasomal components. Interestingly, they also identified novel kinases, the deubiquitinating enzyme otubain 1 and UB ligases (KPC, skip-cullin-F-box and possibly Ube4b). Similarly to proteomic studies of the SNpc, comparison of results between both studies on cortical LBs showed minimal concordance, raising the possibility of biases related to samples or methods differences.

Because the physiological functions of proteins are likely related to their specific intracellular localization, isolation of these specific cellular compartments can be considered as an entry point to understand the cellular mechanisms underlying brain disorders. Organelle proteomics consists in the enrichment and isolation of fractions containing proteins associated with organelles of interest via classical methods such as fractionation, immunoprecipitation or affinity pull-down strategies, among others, prior to proteomic investigation (reviewed in [133–135]). Various subcellular structures such as mitochondria [136,137], plasma membranes [138], synaptic vesicles [139] or post-synaptic structures [140,141] have been already studied using proteomic approaches, and may contribute to understand their role in PD pathogenesis. For example, extensive investigations identified and quantified proteins of mitochondria-enriched fractions isolated from human SNpc of MPTP-treated mice and PD patients compared to age-matched controls using an unbiased quantitative proteomic approach [77,142]. In the latter, by comparing the two groups differentially labelled with ICAT and analysed by a linear trap quadrupole (LTQ) mass spectrometer, authors reported the identification of as much as 842 proteins with 119 displaying significant changes in their relative abundance. Interestingly, mortalin was found again to be substantially and preferentially decreased in the mitochondrial fractions of PD brains as well as in the rotenone-induced cell model of PD. Another study reported a sequential method for the isolation of highly purified NM granules found in DA neurons of human SNpc prior to proteomic analysis [143]. Seventy-two proteins were identified by LCMS/MS analysis, of which the majority was shared with melanosomes and lysosomes. These

results provide important insights into the cellular processes generating the NM granules. As many evidences suggest a functional role for NM in PD neurodegeneration [21,22,144,145], the comparison of NM granules content between control and diseased states may offer some important clues regarding NM toxicity in PD.

Finally, a body of evidence suggests that PTMs may contribute to disease pathology, particularly in the SNpc. Approaches allowing the high throughput proteomic analysis of specifically modified subproteomes such as oxidised, phosphorylated, ubiquitinated or glycosylated [146] proteins have been developed. Until now they were mainly applied to cell lines or animal models and to a lesser extent to human brain tissues of AD or PD patients.

7. Potential avenues and conclusion

After more than a decade of intensive research on α -SYN aggregation, it becomes increasingly clear that α -SYN cannot by itself elucidate all aspects of PD pathogenesis. First, as already mentioned above, α -SYN aggregates are not unique to PD but can be found in a variety of conditions grouped under the label synucleinopathies, a group of enormously heterogeneous conditions with respect to their clinical manifestations, neuroradiological expression and neuropathological features. α -SYN can also be found incidentally in otherwise neurologically intact individuals. Even at the cellular level, α -SYN deposition may occur not only in neurons but also in other cell types, such as oligodendrocytes in multiple system atrophy where they are referred to as glial cytoplasmic inclusions. Second, α -SYN pattern of immunostaining is primarily located in the periphery of mesencephalic LBs and is not prominent in its core as would be expected for a molecule presumed to play an initiating role in seeding aggregation. Conversely, ubiquitinated proteins and components of the UPS are found in the LB centre. Third, based on its intense staining using immunohistochemistry techniques, α -SYN is considered a major if not the essential component of LBs. However, immunostaining methods are notoriously not quantitative and, in fact, a truly quantitative assessment of the many LB components is still awaited. Furthermore, the totality of LB constituents has not been established yet and proteomic studies on cortical LBs [131,132] suggest that α -SYN is just one among hundreds of proteins that compose LBs. Fourth, while numerous studies have focused on α -SYN and its high propensity to aggregate under certain conditions, limited information is available about other proteins that may exhibit similar properties such as parkin, UCH-L1, LRKK2, DJ-1, omi/HtrA2, PINK1 and others. For example, the *in vitro* overexpression of parkin associated with an inhibition of proteasomal activity, was recently shown to lead to the formation of inclusions sharing many characteristics with LBs such as their organisation or immunoreactivity against UB, and some proteasome components [147], supporting the notion that other proteins than α -SYN may induce aggregation. A way to identify aggregation-prone proteins may rely on their primary aminoacid sequence, as proteins containing the hydrophobic aminoacids Ala, Val, Leu, Ile, Phe, Trp, Met or Pro may be more at risk of misfolding or being resistant to refolding. It is well possible that many LB constituents belong to this class of hydrophobic proteins. In addition, LBs may be enriched with natively unfolded proteins, such as α -SYN,

which comprise around 30% of eukaryotic proteins [148] and are naturally prone to aggregation. To verify this hypothesis, a screening of the primary sequences of the identified proteins may be undertaken using algorithms that have been developed to predict these regions of extended disorders [149]. Finally, α -SYN was shown to display a large number of binding partners in cell lines [150]. If α -SYN itself is not toxic when present at physiological concentrations, its interaction with other molecules could lead to neuronal degeneration and, again, the proteomic characterization of these interactants by pull-down assays on LB from post-mortem tissues may be of interest.

Consequently, we propose a reappraisal of the protein aggregation hypothesis that, in our opinion, should go beyond the study of α -SYN alone. We here hypothesize that PD may be a more widespread brain “proteinopathy” (or protein conformational disorder) whereby a number of aggregation-prone neuronal proteins may experience conformational modifications that promote abnormal protein-to-protein interactions and increase binding capacity between related but not necessarily similar, hydrophobic, natively unfolded or partially folded polypeptides. These conformational modifications may result from distinct mechanisms, including overexpression, PTMs or miscleavage of one or a few aggregation-prone proteins, and may initiate a cascade of events by which these proteins may become inactive or toxic. Then, small aggregates formed anywhere within neuron body and processes may coalesce and grow into insoluble, degradation-resistant inclusions. Identifying which proteins are best candidates for these events to occur and deciphering which mechanisms are more likely to be involved, are all crucial issues that may be addressed by means of specific pre-analytical and analytical proteomic strategies.

To achieve this goal, we believe that much effort should be placed on sample purification and fractionation prior to proteomic analyses. For the study of human SNpc, as shown on Fig. 2, we propose an experimental workflow that allows isolating proteomes of interest with an increasing level of tissular, cellular and subcellular specificity. At the tissular level, the mesencephalon from neuropathologically-proven PD cases and matched controls should be carefully and selectively dissected, frozen and cryosectioned. Delineating the whole SNpc proteome and comparing it between the two groups may provide information on global changes occurring in this structure in PD. At the cellular level, a proteomics-compatible number of bromophenol blue stained-DA neurons are extracted from the same sets of samples using LCM, yielding three distinct neuronal populations to be compared, those from healthy subjects and those from PD neurons with and without LBs. Rather than focusing on LBs, this step may be important to assess the whole cellular pathophysiology and particularly to study LB-free nigral degeneration that occur in some genetically-determined forms of PD such as PARK2. At the subcellular level, proteomes of interest from the same sets of neurons can be determined after enrichment of relevant fractions, including nuclei, mitochondria, synapses, NM granules, LBs or other complexes, through subcellular fractionation or immunoprecipitation techniques. It is noteworthy that LCM may not be suitable to purify LBs because of their small size which approaches the capture limits of the LCM technology [132]. Because the same samples are used at each step, comparison of datasets may be less biased by inter-individual

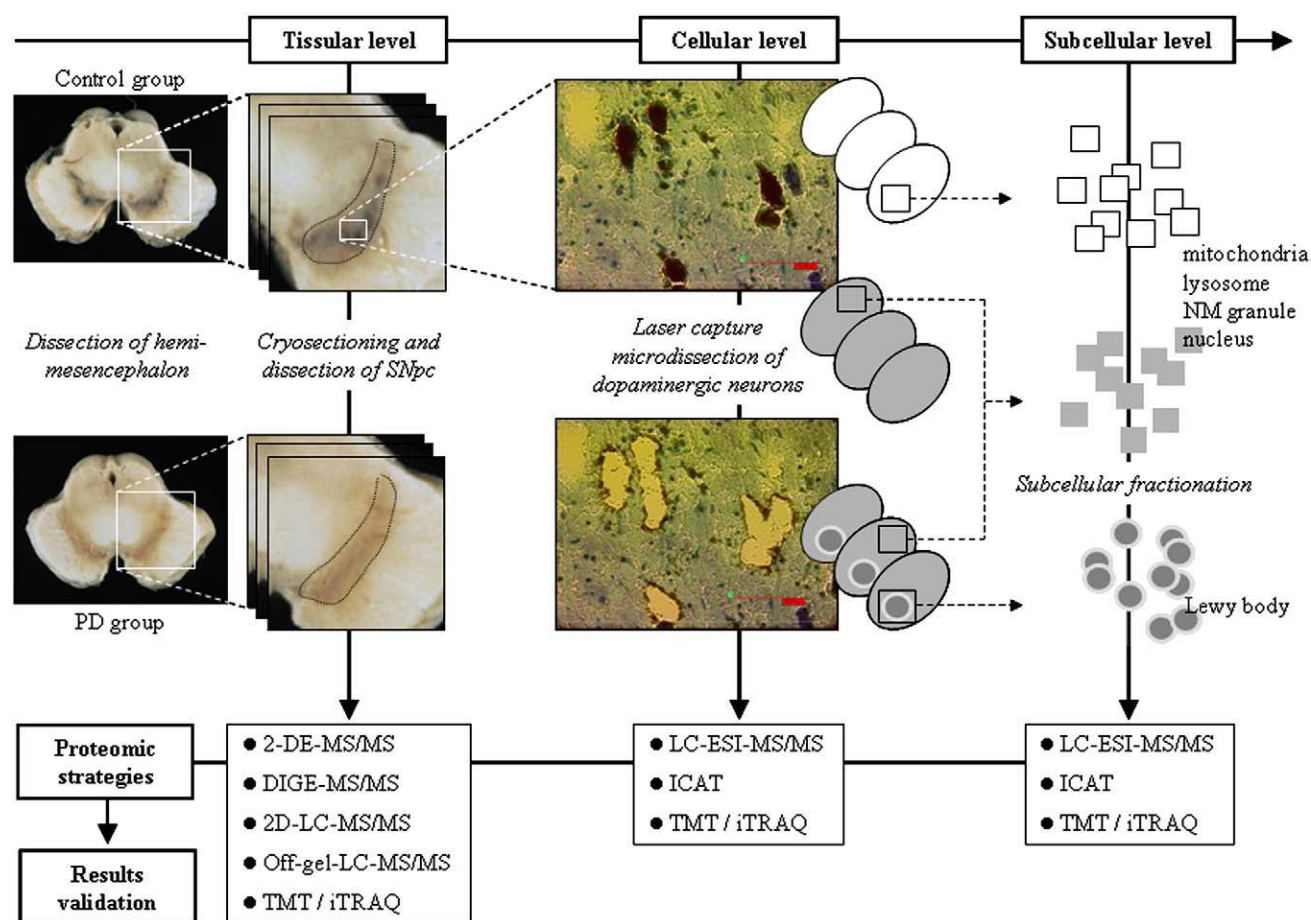


Fig. 2 – Diagram showing a possible proteomic approach to assess biological mechanisms underlying neurodegeneration in the substantia nigra pars compacta from Parkinson's disease human brains obtained at autopsy. At the pre-analytical level, it is proposed to examine samples with an increasing level of specificity at each step and with appropriate proteomics strategies, according to working hypotheses.

variability. Specific proteomic strategies are defined for each level, according to working hypotheses, which involve (1) establishing which proteins may be over- or underexpressed in the SNpc and its components, using quantitative approaches; (2) examining which qualitative protein modifications may play a prominent role in the aggregation processes; and (3) assessing protein mislocalization as another potential mechanism underlying inclusion formation in PD. Data validation is a crucial step that should not be forgotten along the whole procedure.

7.1. Quantitating protein expression

Multiplication of the SNCA gene has been shown to be a potential mechanism by which increased expression of wild-type α -SYN may lead not only to autosomal dominant but also to apparently sporadic PD with LBs. Interestingly, phenotype and disease severity seem correlated with SNCA copy number and probably gene product levels [151]. This observation provides the rationale to study other proteins, whose overexpression and high intracellular concentrations may promote inclusion formation. Quantitative or semi-quantitative methods are highly desirable to examine this hypothesis. Whereas the traditional proteomic approach using 2-DE separation,

comparison of silver-stained patterns of multiple gels and MS/MS identification remains valid, the development of fluorescent and chemiluminescent based technologies has extended its application, allowing the multiplex assessment of protein quantification on a single gel using the DIGE technology and the large scale detection of protein modifications. MS/MS identification should be performed with highly sensitive mass spectrometers that may be able to dissect spots into their sometimes multiple components. Another method called "shotgun proteomics" involves proteins trypsinisation, peptide separations using a broad range of multi-dimensional chromatographic, affinity or gel-based techniques, followed by MS/MS identification. This gel-free alternative has become widespread for relative or absolute quantification using stable isotope labels at a protein or peptide level, and more recently for label-free quantitation strategies. In stable isotope labelling, quantitative peptide ratios — thought to reflect accurately the abundance ratio of the compared proteins — can be obtained from the relative comparison of peptide signal intensities discriminated from single MS or MS/MS scans owing to their mass differences (for reviewed see [152,153]). For example, the isobaric tagging of peptides using iTRAQ or tandem mass tags (TMT) [154] consisting in a reporter and a balance group, enables the

simultaneous protein identification and quantification of up to respectively 6 and 8 samples by MS/MS. After samples pooling, peptides of the same sequence labelled with different versions of a set of multiplex isobaric reagents appear at the same mass in an MS scan, but upon CID fragmentation in the mass spectrometer, give rise to low mass MS/MS signature ions corresponding to the reporter loss. In addition, when proteins of interest are identified, the spiking of a known quantity of an isotope-labelled peptide chemically synthesized as an internal standard can provide absolute quantification [155]. In contrast, label-free quantitation can either be based on the correlation of the MS peak intensities with protein amounts [156] or on the correlation of the number of MS/MS spectra assigned to each protein (spectral counting) [157].

All these quantitative approaches can be used at any step in the workflow proposed on Fig. 2, although we believe gel-based techniques may be insufficiently sensitive and therefore less suitable for cellular or subcellular fraction analyses. To make results more robust, we also propose to use two independent methods of quantification using the same set of samples, in order to identify — and exclude — method-dependent data, and to apply stringent criteria for the determination of differentially expressed spots or MS peaks.

7.2. Examining protein modifications

Proteomic tools allow performing comprehensive study of specific PTMs that are known to occur in the PD SNpc, including the identification and quantification of modified proteins as well as the precise sites of modification.

7.2.1. Oxidation, nitration and S-nitrosylation

In PD, considerable evidence supports oxidative or nitrate stress as key factors involved in neurodegeneration [158]. For example, oxidative modifications of α -SYN via dopamine adducts [159] or nitration [160] facilitate its aggregation. Three assays are commonly used to assess markers of protein oxidation based on the detection of protein carbonyls, tyrosine nitration or cysteine S-nitrosylation.

The first assay involves the reaction of 2,4-dinitrophenylhydrazine with carbonyl groups on proteins, followed by their spectrophotometrical quantification. This technique was applied to SNpc specimens from PD patients and demonstrated increased levels of oxidised proteins compared to controls [161,162]. The high throughput comparison and identification of proteins containing carbonyls in diseased and control brain samples can be determined by an elegant method termed 2DOxyblot, without losing information on protein isoforms. The technique couples 2-DE with the immunohistochemical detection of protein reactive carbonyl groups and finally the matching of 2DOxyblot with 2-DE gels images followed by mass spectrometry of differentially detected spots. This procedure was successfully applied to the comparison of oxidation levels of brain proteins in the inferior parietal lobule of AD subjects versus controls, with the identification of several specifically oxidised proteins, such as creatine kinase BB, glutamine synthase, UCH-L1 and others [163,164].

A second way to index oxidative damage to proteins is through the nitration of tyrosine residues mediated by reactive nitrogen species. Antibodies specific for 3-nitrotyrosine were used to demonstrate increased protein nitrations within LBs in

the melanised neurons of PD brain [165]. The 2DOxyblot technique was also successfully used to map nitrated proteins in AD brains [166]. It is therefore anticipated that the application of 2DOxyblot on human SNpc, DA neurons or subcellular fractions could provide new insights into PD mechanisms.

A third approach has recently emerged from the demonstration that the nitrosylation reaction of cysteine thiol side chain with nitric oxide species forms S-nitrosylated derivatives. This mechanism has recently been linked to unfolded protein response [167] and neurodegeneration in PD [167–169]. The detection of endogenously S-nitrosylated proteins mainly relies on the biotin-switch assay pioneered by Jaffrey and Snyder and is compatible with a broad range of methods such as SDS-PAGE, immunodetection and mass spectrometry [170]. Using this technique, the protein-disulphide isomerase (PDI) chaperone, which assists in the maturation and transport of unfolded secretory proteins was shown to be S-nitrosylated in PD brains. The modification blocked its protective enzymatic activity, leading to the accumulation of polyubiquitinated proteins and activating the unfolded protein response. These data support a previously unrecognized relationship between nitric oxide and protein misfolding in degenerative disorders [167]. Moreover, two studies reported that parkin was S-nitrosylated in various cellular and animal models as well as in sporadic PD brains [168,171]. These authors found that, although overexpression of parkin could rescue neuronal cell lines exposed to proteasome inhibition, the S-nitrosylation could inhibit parkin ubiquitin E3 ligase activity and alter its protective function [168,171]. S-nitrosylation may preclude proper ubiquitination of parkin substrates resulting in their aggregation. These findings provide a molecular link between nitric oxide toxicity, UPS impairment and protein accumulation in sporadic PD.

Lately, a high throughput proteomic approach termed SNOSID (S-Nitrosylation Site Identification) has been developed for the simultaneous identification of proteins containing SNO-cysteine sites in complex biological mixtures [172]. Built on the biotin-switch technique, this innovative method combines the biotinylation of protein SNO-cysteine residues followed by trypsin digestion, affinity purification of biotinylated-peptides, and amino acid sequencing by liquid chromatography tandem MS [172]. To the best of our knowledge, this approach has not yet been used in PD research.

7.2.2. Phosphorylation

Protein phosphorylation is one of the most frequent PTMs and a critical regulatory mechanism of cellular homeostasis including proliferation, gene expression or signal transduction [173–175]. For example, in AD and other tauopathies, a massive increase in the concentration of free hyperphosphorylated tau is thought to be at the heart of abnormal filaments assembly. In synucleinopathies, α -SYN was found to be extensively and selectively phosphorylated at Ser129 in aggregates, as demonstrated by MS and histochemical techniques [176], and this modification seems to promote fibrillogenesis in vitro [176].

A variety of strategies have been introduced for the large scale identification of phospho-proteins and phospho-peptides, the determination of their sites of phosphorylation on Ser, Thr or Tyr residues and their quantitative analysis (see review [177–179]). Quantitative phosphoproteomics is based on 2-DE or MS technologies. The use of phosphorylation-specific stains

Table 3 – Some proteomics-derived proteins of interest for PD research.

SwissProt ID	Protein name	Function	Sample	Quantification method	Observed changes ^a	Hypothetic significance	Reference
P00450	Ceruloplasmin	Iron transporter	CSF	iTRAQ LCMS/MS	↓ (0.65)	Free iron-induced oxidative stress	[101]
P02749	Apolipoprotein H	Transporter of various substrates (i.e. phospholipids, heparin)	CSF	iTRAQ LCMS/MS	↓ (0.66)	Unclear	[101]
P01024	Complement proteins C3 isoform C3c phosphorylated		Serum	2DE	↑ (~7)		[105]
P01024	Complement proteins C3 isoform C3dg	Immune/inflammatory response through complement system activation	Serum	2DE	↑ (~5.5)		
P08603	Complement factor H, short spliced form		Serum	2DE	↑ (~3)		[105]
P54289	Voltage-dependent calcium channel subunit alpha-2/delta-1	Calcium-dependent signal transduction	SNpc	2DE	↑ (2.7)	Increased calcium influx in DA neurons	[108]
P32119	Peroxiredoxin 2	Redox metabolism	SNpc	2DE	↑ (2.0)	Acts against increased oxidative stress	[108]
O75368	SH3 domain binding glutamic acid rich protein like (SH3BGRL)	Redox metabolism	SNpc	2DE	↑ (1.85)	Acts against increased oxidative stress	[109]
O75947	ATP synthase D chain	Energy production	SNpc	2DE	↑ (1.75)	Compensatory mechanism for ATP production inhibition	[108]
P47985	Ubiquinol cytochrome c reductase iron-sulfur subunit	Energy production	SNpc	2DE	2 spots: ↑ (3) and ↑ (1.6)	Compensatory mechanism for ATP production inhibition	[108]
O95299	NADH-ubiquinone 42 kDa subunit	Energy production	SNpc, mitochondrial fraction	ICAT LCMS/MS	↓ (0.34)	Mitochondrial complex I dysfunction	[73]
P33518	Cytochrome c oxidase polypeptide 1	Energy production	SNpc, mitochondrial fraction	ICAT LCMS/MS	↓ (0.69)	Mitochondrial complex I dysfunction	[73]
Q14019	Coactosin-like 1	Cell structure, binds to F-actin	SNpc	2DE	↑ (2.0)	Structural reorganisation	[109]
P09455	Cellular retinol-binding protein 1	Intracellular transport of retinol and retinal	SNpc	2DE	↑ (1.76)	Compensatory mechanism for loss of aldehyde dehydrogenases	[109]
P00352	Aldehyde dehydrogenase 1 A1	Retinoid metabolism, detoxification of aldehydes	SNpc, mitochondrial fraction	ICAT LCMS/MS	↓ (0.38)	Decreased detoxification of aldehydes	[73]
P38646	Mortalin	Chaperone-like protein, control of cell proliferation, cellular aging	SNpc	2DE	↓ (0.75)		[109]
			SNpc, mitochondrial fraction	ICAT LCMS/MS	↓ (0.45)	Reduced capacity to interact with multiple binding partners or to refold proteins	[73]
			Frontal cortex, cytosolic fractions	iTRAQLC/MSMS, 2DE	↓ (0.66–0.49)		[112]

^a In parentheses, PD samples over controls ratio.

such as Pro-Q Diamond fully compatible with other staining methods (e.g., DIGE) and modern MS-based analysis [180] has allowed parallel determination of protein expression changes and altered phosphorylation patterns within a single 2-DE gel experiment [181]. This labor-intensive approach is now supplanted by the development of phospho-proteins/peptides enrichment methods combined with elegant MS identification and quantification using stable isotope labelling [182] or label-free methods [183].

7.2.3. Ubiquitination

Covalent addition of single or multiples units of UB, typically to lysine residues, is a crucial mechanism involved in the targeting of intracellular proteins for 26S proteasomal degradation but also in various other functions [184]. UB is a small, highly conserved protein that modifies substrates through a cascade involving activating (E1), conjugating (E2), ligase (E3) and deubiquitinating enzymes (DUBs) [185]. As human genome may encode a few E1s but around fifty E2s and hundreds of E3s, the scope of protein ubiquitination under pathological conditions could be large [186]. The study of ubiquitinated proteins provides a direct tool to assess the role of the UPS pathway in PD pathogenesis. If protein aggregation and LB formation do originate from an impaired 26S proteasomal function [187] or from some altered DUBs, then the majority of LB components, or at least proteins composing LB fibrils, are expected to be fully ubiquitinated or, conversely, deubiquitinated by DUBs following aggregation. This does not seem to be the case in PD but this issue requires further investigation. The determination of the precise form of UB in aggregates and its substrates might help in improving our understanding of the process. Strategies have been recently developed for the affinity purification of UB-conjugates, including UB antibodies [188] and UB binding proteins [189]. However, methodologies still need to be refined to allow the large scale routine analysis of these modifications. Given the numbers of multi-protein complexes involved in the UPS machinery and all the substrates susceptible to be targeted, large scale proteomic approaches are necessary to fully understand the role of ubiquitination in diseases in general, and in PD in particular.

7.3. Studying protein miscleavage

The role of proteolytic cleavage in degenerative diseases has been established in AD where the abnormal involvement of sequentially active secretases leads to the accumulation and aggregation of pathogenic A β ₄₂ and the formation of amyloid plaques. Moreover, inclusions in post-mortem tissues of patients with Huntington's disease contain short N-terminal fragments of polyglutamine repeats [190], that seem to be more toxic than full-length huntingtin [191,192]. In PD, the role of proteolytic products is less clearly established. C-terminally truncated α -SYN seems to be present in LBs of patients with dementia with LBs [193] and more prone to aggregate in vitro [160,194,195]. Moreover, evidences suggest that the NAC peptide of α -SYN identified in dementia with LBs could also aggregate as toxic entities [196]. The development of a novel exciting field in proteomics, peptidomics, dedicated to the analysis of endogenous peptides that are usually missed by standard gel-based or shotgun proteomic strategies (see [197–199]), could be a valuable

tool in assessing the role of proteolysis in PD neurodegeneration, as recently shown by Nilsson et al. [200].

As summarized in Table 3, proteomic studies performed thus far in the field of PD have already generated a large amount of data that support existing pathogenic theories and have produced a number of original findings regarding PD pathogenesis and biomarkers. However, these developments remain confidential in the neuroscience community and are rarely incorporated into the current scenarios of molecular mechanisms underlying PD. Reasons may be found in some limitations that tend to affect existing clinical (and basic) proteomic studies [201], notably (1) the lack of well-defined and specific working hypothesis preceding the design of the proteomic protocol, which has been used essentially as a screening tool; (2) the choice of samples which may not be adequate for the purpose of the study; (3) an enormous variability of pre-analytical and analytical methods which may preclude inter-laboratory comparisons; (4) the limited number and heterogeneity of samples examined and low statistical power of comparative analyses; (5) the absence of result validation using independent, standardized and reliable methods. These caveats have to be kept in mind when assessing available studies performed in PD research and need to be overcome in future proteomic studies. Based on the literature reviewed above, it could be concluded that the mechanisms at the basis of PD pathogenesis are exceedingly complex and our current understanding of them remains limited. The prevailing theory involves an impairment of selected groups of neurons to handle abnormally processed cellular proteins which deposit as insoluble and toxic aggregates, the cause of which being still elusive. Proteomics-oriented studies published so far tend to support this view and, in fact, suggest that proteomics may be one of the most promising research avenues in the field of PD pathogenesis.

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